

Defecation Reflex + Hirschsprung Disease

Headings

- ▼ Defecation Reflex + Hirschsprung Disease
 - ▼ 1. Defecation Reflex
 - Stimulus**
 - Reflex pathway**
 - Mechanism**
 - ▼ 2. Hirschsprung Disease
 - Definition**
 - Pathophysiology**
 - Clinical Features**
 - Treatment**
 - ★ Last-minute recall

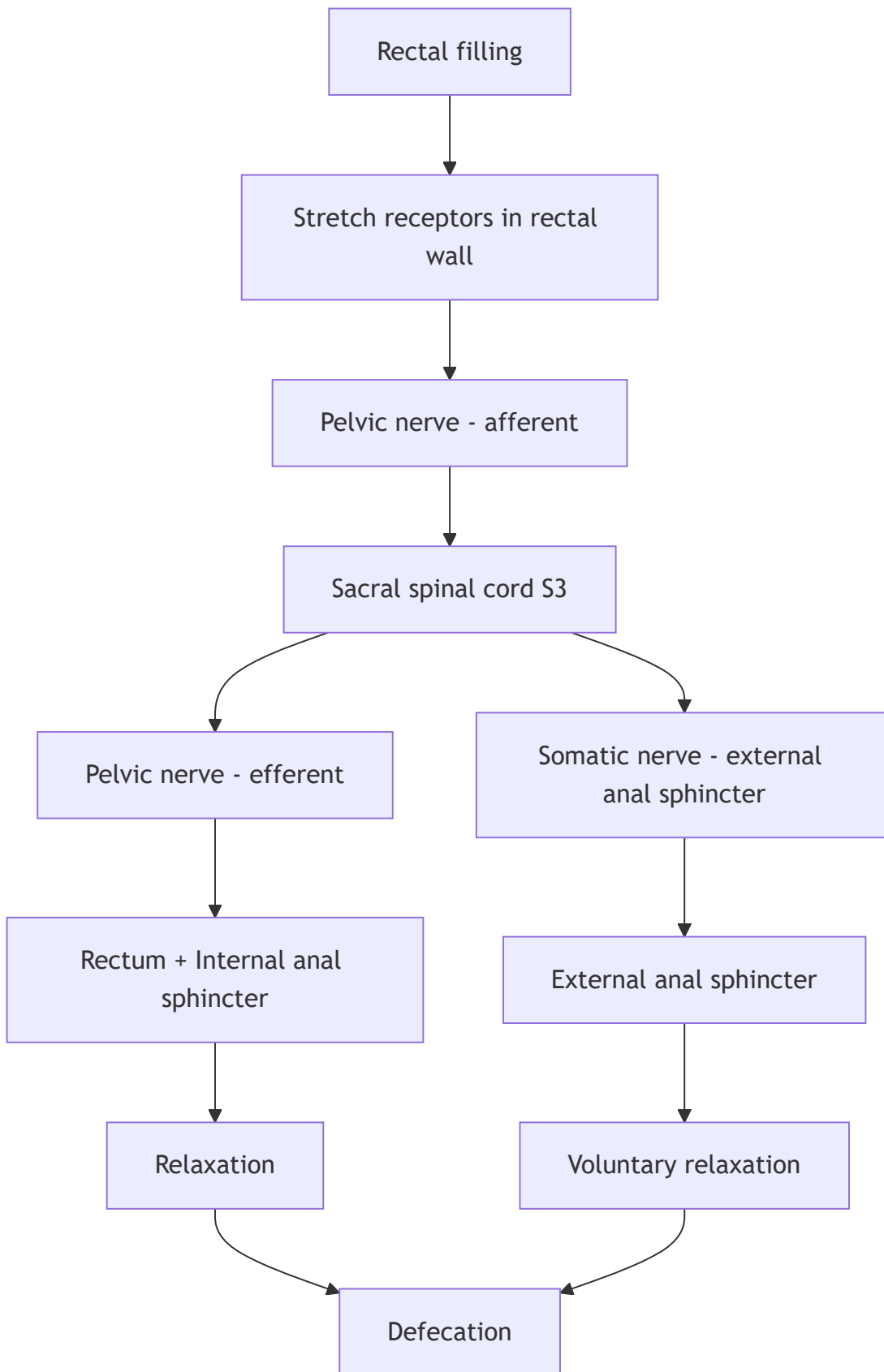
1. Defecation Reflex

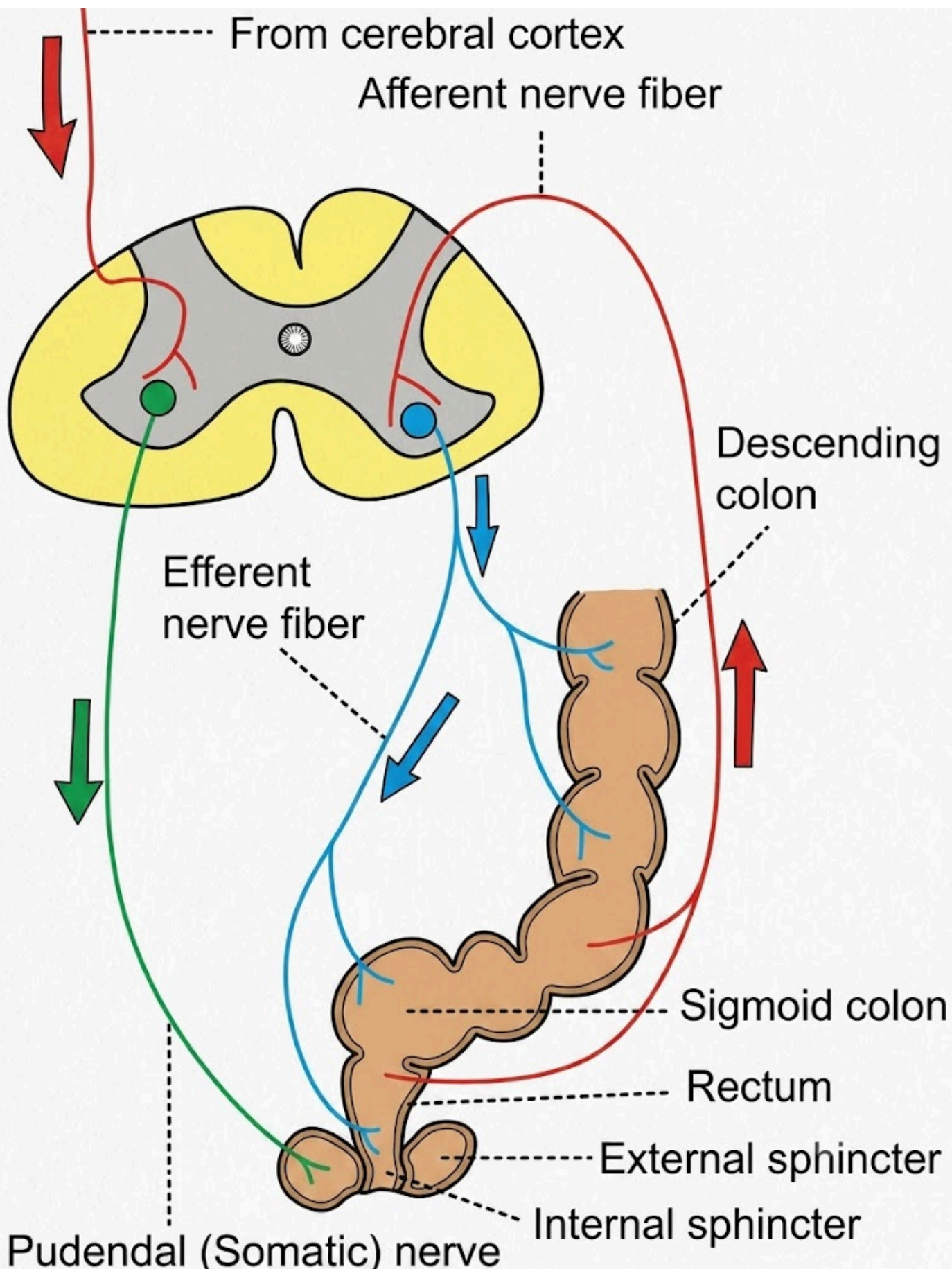
Defecation = reflex phenomenon under voluntary control

Stimulus

Mass peristalsis of descending + sigmoid colon → Rectal filling → Stretch of rectal wall

Reflex pathway





Mechanism

Event	Effect
Rectal distension	Initiates defecation reflex
Myenteric inhibitory signals	Relax internal anal sphincter
Voluntary cortical control	Relaxes external anal sphincter
Puborectalis relaxation	Decreases anorectal angle
Deep inspiration + diaphragm contraction	↑ Intra-abdominal pressure
Abdominal muscle contraction	Further ↑ intra-abdominal pressure
Straining	Forces feces through anal canal

↑ Intra-abdominal pressure; +; Sphincter relaxation → Defecation

Normal anorectal angle: ~90°

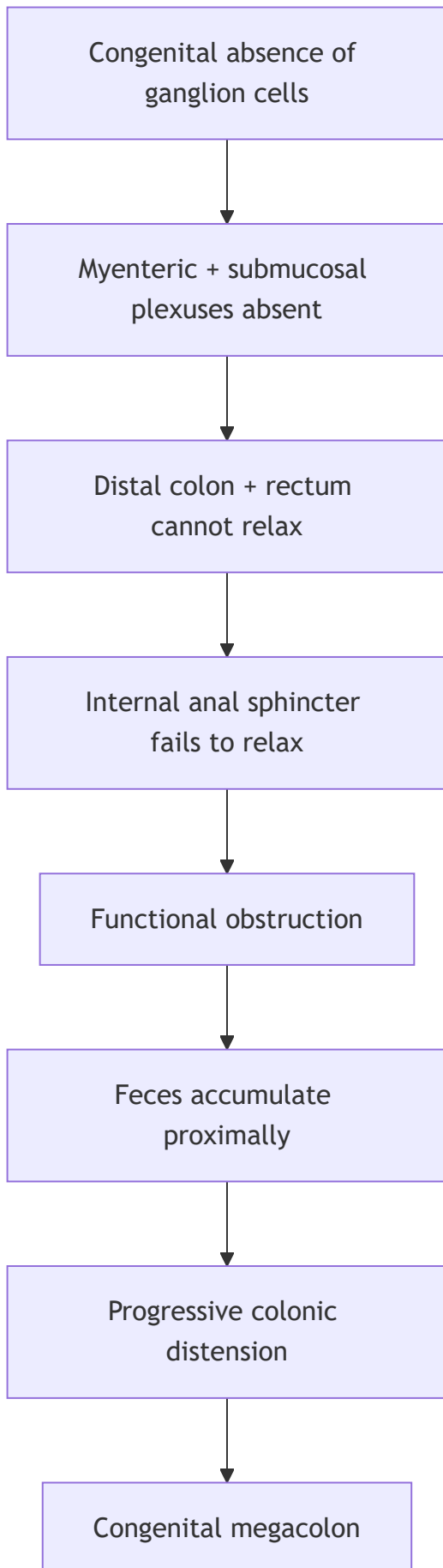
Applied: Defecation is fundamentally a **spinal reflex**; higher centres, especially the cortex, exert voluntary control.

2. Hirschsprung Disease

Definition

Hirschsprung disease = congenital megacolon due to aganglionosis.

Pathophysiology



Clinical Features

- Abdominal distension
- Anorexia
- Lassitude
- Severe cases → symptoms may appear in the **newborn period**

Treatment

→ **Surgical treatment** to remove/bypass the aganglionic segment and restore bowel passage.

★ Last-minute recall

Rectal filling → Stretch receptors → S_3 → Pelvic nerve → Internal sphincter relaxation

Hirschsprung

No ganglion cells → No relaxation → Obstruction → Megacolon